1. Overview

This report documents analysis of large-scale anisotropy in the mean redshift field ⟨z⟩ of the Quaia quasar catalogue ($G < 20$). The goals are:

- Measure the dipole of ⟨z⟩ in different Galactic latitude cuts ($|b| > b\_cut$) and redshift slices.
- Compare the measured dipoles with the CMB kinematic dipole to look for parallel or anti-parallel alignment.
- Use rotation Monte Carlo (MC) to estimate the significance of any observed alignment.
- Recently: increase the minimum number of objects per pixel from 3 to 5 (N_MIN_PER_PIX = 5) to improve robustness, then re-run the full rotation MC suite.

There is emerging evidence for a **redshift-modulated alignment pattern**:

- mid-z bins (roughly 0.5–1.0) tend to show **CMB-parallel** dipoles;
- high-z bins (1.5–2.5) tend to show **CMB-anti-parallel** dipoles;
- one mid-z bin (0.50–0.75) has an **unusually large dipole amplitude**, independent of CMB alignment.

At present, the statistical significance is modest (typically $\leq 2\sigma$), but the pattern is persistent across masks and the new $N \geq 5$ maps, and is therefore worthy of further investigation.

Note: This report has a number of report updates that may look a bit disjointed due to the addition of forthcoming results from various programs and analysis options.

Re-Indexing has not been addressed

At this stage HHT has not been actioned.

Planning on enhancing z slices to 0.2 in order to produce more bins and possibly modify min pixel count to be adaptive.

2. Data, maps and masks

- **Catalogue:** Quaia G20.0 (755,850 objects).
- **Magnitude cut:** G < 20.
- **HEALPix resolution:** NSIDE = 64 (npix = 49,152).

Galactic latitude cuts

$$/b/>b_{cut}, b_{cut}=10^{\circ}, 15^{\circ}, 20^{\circ}, 25^{\circ}, 30^{\circ}, 35^{\circ}.$$

Additional Galactic centre (GC) hole

- centre (l, b) = (0°, 20°), radius 40°.

Minimum per pixel (new setting)

$$N_{min}=N_{MIN\_PER\_PIX}=5$$

i.e. only pixels containing at least 5 quasars contribute to the ⟨z⟩ dipole fits.

Redshift slices currently used

Non-overlapping slices:

- $0.10 \le z < 0.30$
- $0.30 \le z < 0.50$
- $0.50 \le z < 0.75$
- $0.75 \le z < 1.00$
- $1.00 \le z < 1.50$
- $1.50 \le z < 2.50$

Each slice is tagged in the code as:

- z0p10_0p30,
- z0p30_0p50,
- z0p50_0p75,
- z0p75_1p00,
- z1p00_1p50,
- z1p50_2p50.

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3. Dipole model and definitions

Let ⟨z⟩(n̂) be the mean redshift in direction n̂ on the sky.

We fit a monopole + dipole model

$$\langle z \rangle(\hat{n}) = z_0 + A \cdot \hat{n},$$

where

- $z_0$is the monopole,
- $A$is the dipole vector with amplitude

$$A = / A /.$$

The best-fit direction is expressed in Galactic coordinates (l, b).

The CMB dipole direction is denoted $\hat{d}_{CMB}$.

We project the ⟨z⟩ dipole onto this direction:

$$A_{/\!/} = A \cdot \hat{d}_{CMB},$$

and define the **CMB-parallel fraction**

$$f_{/\!/} = \frac{A_{/\!/}}{A}.$$

Interpretation:

- $f_{/\!/} \approx +1$: dipole almost parallel to CMB dipole.
- $f_{/\!/} \approx 0$: roughly orthogonal.
- $f_{/\!/} \approx -1$: almost perfectly anti-parallel.

We also measure the angular separation between the ⟨z⟩ dipole direction and the CMB dipole:

$$\theta_{sep} = \arccos\left(\hat{A} \cdot \hat{d}_{CMB}\right), \hat{A} = \frac{A}{/ A /}.$$

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4. Rotation Monte Carlo (MC) method

For each $\left(b_{cut}, \text{z-slice}\right)$ combination:

1. Take the observed ⟨z⟩ map (with mask and **N ≥ 5** per-pixel cut).

2. Generate $n_{mock} = 2000$ random rotations of the entire map on the sphere (preserving the sky pattern but scrambling its orientation relative to the CMB dipole).

3. For each rotated map, recompute the dipole:
    - amplitude $A^{(i)}$,
    - CMB-parallel component $A_{/\!/}^{(i)}$,
    - fraction $f_{/\!/}^{(i)}$.

4. Compute empirical p-values:
    - for the amplitude

$$p(A) = \frac{\#\{i : A^{(i)} \geq A_{obs}\}}{N_{mock}},$$

    - for alignment

$$p(/A_{/\!/}/) = \frac{\#\{i : /A_{/\!/}^{(i)}/ \geq /A_{/\!/,obs}/\}}{N_{mock}},$$

$$p(/f_{/\!/}/) = \frac{\#\{i : /f_{/\!/}^{(i)}/ \geq /f_{/\!/,obs}/\}}{N_{mock}}.$$

These are effectively two-sided tail probabilities relative to the rotation ensemble.

Rough σ-equivalents:

- $p \approx 0.32 \rightarrow \sim 1\sigma$
- $p \approx 0.10 \rightarrow \sim 1.6\sigma$
- $p \approx 0.05 \rightarrow \sim 1.96\sigma$

No single bin currently exceeds ≈ 2σ, but some are in the 1.5–2σ range, especially at high z and for certain latitude cuts.

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5. Effect of raising N_MIN_PER_PIX from 3 to 5

We rebuilt the ⟨z⟩ maps and dipole grid with **N_MIN_PER_PIX = 5**, then re-ran the full rotation MC suite.

Key consequences

- **Low-z slice 0.10–0.30:**
  - N_{good}pixels with N ≥ 5 collapses to only 1–3 pixels across all |b| cuts.
  - This slice is not usable for robust dipole estimation with N ≥ 5.
  - Any apparent “strong” low-z signal from earlier N ≥ 3 runs must therefore be treated as fragile and possibly dominated by a few noisy pixels.
- **Mid- and high-z slices (z ≥ 0.30):**
  - Still have hundreds to tens of thousands of good pixels.
  - Dipole measurements remain stable and well-constrained.

Summary

The N ≥ 5 cut substantially improves robustness at the cost of discarding the shallow 0.10–0.30 slice.

All current interpretations therefore focus on **z ≥ 0.30** and on the new **N ≥ 5 rotation MC results**.

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6. Current dipole grid: qualitative summary (N ≥ 5)

From `quaia_dipole_bcut_grid.txt` (N_MIN_PER_PIX = 5, NSIDE 64):

6.1 Full sample (all z, varying |b| cuts)

- Amplitudes $A \approx (4.6 - 5.7) \times 10^{-3}$ across $b_{cut} = 10 - 35^{\circ}$.
- Dipole directions are roughly in the Galactic anti-centre hemisphere, with latitudes around b ≈ 0° to −16°.
- Angular separations from the CMB dipole are large, $\theta_{sep} \approx 85 - 100^{\circ}$, i.e. approximately orthogonal to the CMB dipole.
- Corresponding fractions $f_{\parallel}$ are small: $-0.18 \lesssim f_{\parallel} \lesssim +0.06$.

So the **global ⟨z⟩ dipole does not align strongly with the CMB dipole.**

6.2 Redshift dependence (example: |b| > 20° cut)

Below are the notable trends for |b| > 20° (numbers approximate):

- **0.30 ≤ z < 0.50**
    - $A \approx 3.4 \times 10^{-3}$, $N_{good} \approx 180$.
    - Dipole direction roughly at (l ≈ 87°, b ≈ 42°).
    - Separation from CMB: $\theta_{sep} \approx 90^{\circ}$.
    - $f_{\parallel} \approx 0$ (nearly orthogonal).
- **0.50 ≤ z < 0.75**
    - $A \approx 4.5 \times 10^{-3}$.
    - Direction (l ≈ 128°, b ≈ 43°).
    - $\theta_{sep} \approx 82^{\circ}$ (still roughly orthogonal, but now in the same quadrant as CMB).
    - $f_{\parallel} \approx +0.15$ (mildly CMB-parallel).
- **0.75 ≤ z < 1.00**
    - $A \approx 1.1 \times 10^{-3}$.
    - Direction (l ≈ 114°, b ≈ 69°).
    - $\theta_{sep} \approx 61^{\circ}$.
    - $f_{\parallel} \approx +0.48$ → **moderately CMB-parallel.**
- **1.00 ≤ z < 1.50**
    - $A \approx 2.0 \times 10^{-3}$.
    - Direction (l ≈ 60°, b ≈ 13°).
    - $\theta_{sep} \approx 116^{\circ}$.
    - $f_{\parallel} \approx -0.43$ → **moderately anti-parallel.**

- **1.50 ≤ z < 2.50**
  - $A \approx 2.1 \times 10^{-3}$.
  - Direction (l ≈ 90°, b ≈ −25°).
  - $\theta_{sep} \approx 157^{\circ}$.
  - $f_{//} \approx -0.92$→ **strongly CMB-anti-parallel.**

This is the key pattern:

Mid-z bins (~0.5–1.0) tend to align **parallel** to the CMB dipole, whereas high-z bins (≥ 1.5, especially 1.5–2.5) tend to align **anti-parallel** to the CMB.

The same qualitative behaviour persists across $b_{cut} = 15^{\circ},\ 20^{\circ},\ 25^{\circ},\ 30^{\circ},\ 35^{\circ}$:

- low-to-mid z: $f_{//} > 0$(CMB-parallel),
- high z: $f_{//} < 0$(almost anti-parallel).

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7. Selected rotation MC results (N ≥ 5 runs)

The latest rotation MC run (N_MIN_PER_PIX = 5, n_mock = 2000, seed = 42) gives updated p-values that are fully consistent with the current pixel cut.

7.1 Slice 0.50 ≤ z < 0.75: strong dipole amplitude, weak CMB link

This bin shows the **clearest amplitude anomaly**, but no strong preference for CMB alignment.

- |b|>10°:
  - $A_{obs} \approx 3.82 \times 10^{-3}$,
  - $f_{//,obs} \approx +0.40$.
  - $p(A) = 0.001 \rightarrow$ **very rare amplitude in rotations**.
  - $p(/f_{//}/) \approx 0.61 \rightarrow$ alignment not unusual.

- |b|>15°: $p(A) = 0.045$.
- |b|>20°: $p(A) = 0.009$.
- |b|>25°: $p(A) = 0.002$.
- |b|>30°: $p(A) = 0.083$.
- |b|>35°: $p(A) = 0.049$.

Across these cuts, the dipole amplitude is **consistently high** relative to the rotated skies (p(A) typically 0.001–0.05), while $p(/f_{//}/)$stays large (~0.65–0.70).

This suggests a **real, strong ⟨z⟩ dipole in the 0.50–0.75 bin**, but not one that is specifically tied to the CMB dipole direction.

7.2 Slice 0.75 ≤ z < 1.00: CMB-parallel tendency at high |b|

Amplitudes in this slice are modest, but at high latitude cuts the dipole becomes **strongly CMB-parallel**:

- |b|>30°:
  - $f_{//,obs} \approx +0.90$.
  - $p(/f_{//}/) \approx 0.095$(~1.7σ).

- |b|>35°:
  - $f_{//,obs} \approx +0.94$.
  - $p(/f_{//}/) \approx 0.064$(~1.9σ).

Amplitude p-values in this slice are large (p(A) ≈ 0.18–1.0), i.e. unremarkable in size.

The interest here is purely **directional**: a marginal (~1.7–1.9σ) preference for **CMB-parallel alignment** at 0.75–1.00 when restricting to high latitudes.

7.3 Slice 1.50 ≤ z < 2.50: persistent CMB-anti-parallel signal

This slice shows the most consistent CMB-anti-parallel behaviour:

- |b|>10°:
  - $f_{\parallel, obs} \approx -0.90$,
  - $p(/f_{\parallel}/) = 0.10$.

- |b|>15°:
  - $f_{\parallel, obs} \approx -0.95$,
  - $p(/f_{\parallel}/) = 0.045 (\sim 2\sigma)$.

- |b|>20°:
  - $f_{\parallel, obs} \approx -0.92$,
  - $p(/f_{\parallel}/) = 0.082$.

- |b|>25°:
  - $f_{\parallel, obs} \approx -0.85$,
  - $p(/f_{\parallel}/) = 0.141$.

- |b|>30°:
  - $f_{\parallel, obs} \approx -0.97$,
  - $p(/f_{\parallel}/) = 0.023$(**most extreme case, ~2.0–2.3σ**).

- |b|>35°:
  - $f_{\parallel, obs} \approx -0.84$,
  - $p(/f_{\parallel}/) = 0.145$.

Amplitude p(A) values here are mostly unremarkable (0.35–0.74, except p(A)=0.038 for |b|>30°).

The primary anomaly in this slice is **directional**: a **high-z dipole consistently anti-parallel to the CMB**, with the strongest case (|b|>30°) at **p(|f_\parallel|) ≈ 0.023**.

7.4 Full-sample bins

For the **full-z** sample at each |b| cut:

- $p(A) \approx 0.26 - 0.98$,

- $p(/f_{\parallel}/) \approx 0.81 - 0.97$.

The full-sample dipole amplitude and orientation are **completely consistent with the rotation ensemble**, i.e. no single all-z kinematic-like anomaly.

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8. Possible redshift-modulated alignment ("modulation artefact")

The current results, especially from `quaia_dipole_bcut_grid.txt` and the updated $N \geq 5$ rotation MC, suggest a systematic evolution of $f_{//}$ with redshift:

- $f_{//}$ transitions from **positive (CMB-parallel)** at mid-z to **strongly negative (CMB-anti-parallel)** at high-z.
- This pattern is present across multiple |b| cuts, indicating that it is not simply a local mask artefact.
- The new MC confirms that several of these bins have p-values in the 0.02–0.10 range (roughly 1.6–2.3σ), particularly:
  - amplitude in 0.50–0.75,
  - CMB-parallel alignment in 0.75–1.00 at |b|≥30–35°,
  - CMB-anti-parallel alignment in 1.50–2.50 at |b|≥15°, especially |b|>30°.

There are at least three broad possibilities:

1. **Astrophysical / cosmological modulation**
   Large-scale structure and peculiar motions could produce a redshift-dependent dipole pattern that changes phase relative to the CMB rest frame.

   More exotic interpretations could involve evolving anisotropy or foreground structures.

2. **Survey / selection systematics**
   Redshift-dependent completeness, extinction residuals, or calibration drifts across the sky could imprint a spurious dipole whose direction varies with z.

   Masking and variable depth near the Galactic plane could contribute.

3. **Statistical fluctuations**
   Given multiple bins and cuts, some 1.5–2σ features are expected by chance.

At this stage, the main novel observation is:

A **redshift-modulated sign flip** in the CMB-parallel fraction $f_{//}(z)$, with mid-z slices mildly parallel to the CMB dipole and high-z slices strongly anti-parallel, plus one mid-z bin with an anomalously **large dipole amplitude**.

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9. Proposed strategy: modulation analysis with sliding z-windows and HHT

To better understand whether this is a real modulation or just a few noisy bins, we can move beyond disjoint slices and treat $f_{//}$(or related quantities) as a function of redshift.

9.1 Sliding redshift windows

Instead of fixed non-overlapping bins, define overlapping windows, e.g. width Δz ≈ 0.4–0.5, step δz ≈ 0.1–0.2, such as:

- 0.10–0.50, 0.20–0.60, 0.30–0.70, …, 1.50–2.50.

For each window:

- Build ⟨z⟩ map with **N ≥ 5**.
- Fit dipole and compute $A, A_{//}, f_{//}$.
- Optionally record $\theta_{sep}$and the raw dipole components in Cartesian form.

This yields a quasi-continuous function of redshift:

- $f_{//}(z_{mid})$, $A(z_{mid})$, $A_{//}(z_{mid})$.

9.2 HHT / EMD-based modulation analysis

Once we have $f_{//}(z_{mid})$sampled at many z_mid values, we can treat it as a 1-D "time series" in z and apply Hilbert–Huang Transform (HHT) methods already used in the HHT/BAO work:

- **Empirical Mode Decomposition (EMD)**
  Decompose $f_{//}(z)$into Intrinsic Mode Functions (IMFs) plus a trend.

  Identify whether there is a dominant mode representing a slow modulation (e.g. transition from + to −).

- **Hilbert spectral analysis**
  For each IMF, compute instantaneous amplitude and phase vs z.

  Look specifically for phase-coherent modes across b_cuts or different catalogue subsets (e.g. NGC vs SGC).

- **Surrogate tests**
  Generate surrogates by randomly scrambling z or by rotating sky positions within each z window, then recompute the $f_{//}(z)$trajectory.

  Apply the same HHT pipeline to surrogates.

  Quantify how often a modulation as strong/coherent as the observed one appears by chance.

- **Cross-comparisons**
  Compare the modulation pattern between different |b| cuts and between NGC/SGC hemispheres.

  A truly cosmological modulation should be robust across reasonable mask changes.

This would let us move from “a few 1.5–2.3σ bins” to a **global significance test of the entire $f_{/\!/}(z)$ pattern.**

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10. Current status summary

- Maps and dipole grid rebuilt with **N_MIN_PER_PIX = 5** (NSIDE 64, G < 20).
- Low-z 0.10–0.30 slice effectively unusable (1–3 good pixels only).
  We now treat any apparent low-z alignment as unreliable.
- Mid- and high-z slices remain robust, with hundreds to tens of thousands of good pixels.
- Dipole grid shows a clear redshift trend in CMB-parallel fraction:
  - mid-z (0.5–1.0): $f_{/\!/} > 0$, roughly CMB-parallel;
  - high-z (≥ 1.5): $f_{/\!/} \approx -0.8$to −0.95, strongly anti-parallel.
- **Updated rotation MC (N ≥ 5, n_mock = 2000):**
  - Full-sample amplitudes and orientations are unremarkable (p(A) ≈ 0.26–0.98, p(|f_\parallel|) ≈ 0.81–0.97).
  - The 0.50–0.75 slice shows a **strong amplitude anomaly** with p(A) typically in the 0.001–0.05 range across |b| cuts, but with no special CMB alignment.
  - 0.75–1.00 shows **CMB-parallel alignment** at high |b|, with p(|f_\parallel|) ≈ 0.06–0.10 (≈1.7–1.9σ).
  - 1.50–2.50 shows **CMB-anti-parallel alignment**, most strongly for |b|>30° where p(|f_\parallel|) ≈ 0.023 (~2.0–2.3σ).
- Overall, several bins now sit in the **p ≈ 0.02–0.10** range (roughly 1.6–2.3σ), but no single result yet exceeds ≈2.5σ.
- **Next major step:** implement a sliding-window, HHT-based modulation pipeline for $f_{/\!/}(z)$to test whether the apparent sign flip and modulation with redshift is statistically meaningful or a chance/systematic artefact.

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11. Acronyms & notation (for quick reference)

- **CMB** – Cosmic Microwave Background.
- **GC** – Galactic Centre.
- **HEALPix** – Hierarchical Equal Area isoLatitude Pixelization.
- **NSIDE** – HEALPix resolution parameter.
- **MC** – Monte Carlo.
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- **EMD** – Empirical Mode Decomposition.
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- **HHT** – Hilbert–Huang Transform.
- **IMF** – Intrinsic Mode Function (in HHT/EMD).
- **N_MIN_PER_PIX (N_min)** – minimum number of objects per pixel required to include that pixel in dipole fits.
- **A** – dipole amplitude of ⟨z⟩.
- **A_∥** – component of the dipole parallel to the CMB dipole.
- **f_∥** – CMB-parallel fraction of the dipole, $f_{//} = A_{//} / A$.
- **θ_sep** – angular separation between the ⟨z⟩ dipole and the CMB dipole.

1. What the shuffle MC with $N \geq 5$ is saying

Big picture: **nothing looks crazy under pixel–shuffle**, even in the bins that looked a bit hot in the rotation MC.

Key points from your log (N_MIN_PER_PIX = 5, n_mock = 2000):

- **Low-z (0.10–0.30)**
  - |b|>10° and 15°: p(A) ≈ 0.33–0.34, p(|A∥|) ≈ 0.49–0.50.
  - But this slice only has N_good = 3 pixels, so it's not trustworthy anyway.
- **0.30–0.50** (all b-cuts)
  - p(A) ≈ 0.64, 0.80, 0.87, 0.66, 0.98, 0.98 → completely unremarkable.
  - No sign of an amplitude excess; any alignment/anti-alignment we saw in rotation MC is purely an orientation effect.
- **0.50–0.75** (the band where rotation MC gave low p(A) for some b-cuts)
  - p(A) ≈ 0.26, 0.27, 0.17, 0.17, 0.34, 0.46 across b_cuts = 10–35°.
  - So under shuffle, the amplitude is **entirely consistent with isotropic noise + mask**.
  - Conclusion: the "low p(A)" from rotation MC here was **not** an intrinsic ⟨z⟩–dipole excess, just how that fixed pattern happens to sit inside the rotating mask.
- **1.00–1.50** (this is where something mildly interesting appears)
  - p(A) by b_cut:
    - |b|>10°: 0.116
    - |b|>15°: 0.048
    - |b|>20°: 0.137
    - |b|>25°: 0.053
    - |b|>30°: 0.083
    - |b|>35°: 0.030
  - So there's a **consistent trend of slightly low p(A)** for 1.0–1.5, especially at higher b_cuts (0.048, 0.053, 0.030 → ~1.7–2.1σ at best, before trials factors).
  - p(|A∥|) are all ~0.14–0.89 → no strong CMB–direction anomaly here; it's an amplitude effect.
- **1.50–2.50** (where rotation MC showed strong anti-parallel f∥)
  - p(A) ≈ 0.56, 0.51, 0.41, 0.34, 0.78, 0.77 across b_cuts.
  - p(|A∥|) ≈ 0.20, 0.13, 0.12, 0.11, 0.32, 0.38.

- So shuffle sees **nothing special in amplitude**; and A∥ along the CMB axis is also not particularly extreme.
- The high-z “anti-parallel” behaviour therefore looks like a **pure orientation effect** relative to the CMB, not extra dipole power.

So the main update is:

Only the 1.0–1.5 slice shows a mild, b_cut-robust excess in dipole amplitude under pixel–shuffle (p(A) ≈ 0.03–0.05 at high |b|).

Everything else, including the 0.5–0.75 band and the high-z 1.5–2.5 anti-alignment, is consistent with an isotropic ⟨z⟩ field once you account for the mask.

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2. Text you can paste into the report (replace / extend sections)

You can drop this in as a **replacement for your current Section 7** and tweak Section 10.

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7. Rotation vs shuffle MC results (N ≥ 5 maps)

With the ⟨z⟩ maps rebuilt at NSIDE = 64 and **N_MIN_PER_PIX = 5**, we have now run **both**:

- the **rotation MC** (quaia_mc_bcut_rotate.py), and
- the **pixel–shuffle MC** (quaia_mc_bcut_shuffle.py, n_mock = 2000, seed = 52).

These probe complementary nulls:

- **Rotation MC** keeps the ⟨z⟩ pattern fixed on the sky and randomly rotates it relative to the CMB dipole → tests whether the observed dipole is **unusually aligned or anti-aligned with the CMB** given the mask.
- **Shuffle MC** keeps the mask and pixel sampling fixed, but **randomly reassigns redshifts**, thus erasing any real large-scale structure → tests whether the **dipole amplitude itself is larger than expected from shot noise + mask**.

All quoted p-values below correspond to the new **N ≥ 5** configuration.

7.1 Low-z, 0.10 ≤ z < 0.30

For all b_cuts (|b| > 10°, 15°, …, 35°) we have **N_good ≈ 1–3 pixels** only.

Shuffle MC gives p(A) ≈ 0.33–0.34 and p(|A∥|) ≈ 0.49–0.50, i.e. nothing unusual under the null, but in practice this slice is **not usable** and we discard it from interpretation.

7.2 0.30 ≤ z < 0.50

Across all b_cuts:

- Shuffle MC: p(A) ≈ 0.64–0.98, p(|A∥|) ≈ 0.75–0.94.

Thus there is **no evidence for an anomalous ⟨z⟩ dipole amplitude** in this band.

Any interesting behaviour seen in rotation MC (e.g. apparent anti-alignment at some b_cuts) is a **pure orientation effect relative to the CMB**, not an excess of dipole power.

7.3 0.50 ≤ z < 0.75

This band previously showed low p(A) in the rotation MC for some b_cuts (particularly |b| > 10°–25°), suggesting a potentially large dipole amplitude. However:

- Shuffle MC now gives p(A) ≈ 0.26 (|b|>10°), 0.27 (15°), 0.17 (20°), 0.17 (25°), 0.34 (30°), 0.46 (35°).
- All corresponding p(|A∥|) are ≈ 0.40–0.72.

So **under the isotropy+noise null, this amplitude is entirely consistent with chance**.

The low p(A) from rotation MC was therefore driven by how the fixed, real dipole pattern sits within the rotated mask, rather than by a genuinely "too big" ⟨z⟩ dipole.

7.4 0.75 ≤ z < 1.00

- Shuffle MC: p(A) ≈ 0.77–0.89 and p(|A∥|) ≈ 0.61–0.66 for |b|>10°–25°, and still high at larger b_cuts.

So there is **no amplitude anomaly**.

The moderate positive f∥ seen in the dipole grid and rotation MC (CMB-parallel tendency) is once again an **orientation feature only**, with amplitude well within expectations.

7.5 1.00 ≤ z < 1.50 – mild amplitude excess

This slice is the **only one where shuffle MC suggests a mildly enhanced ⟨z⟩–dipole amplitude**:

- Shuffle p(A) by b_cut:
    - |b|>10°: 0.116
    - |b|>15°: 0.048
    - |b|>20°: 0.137
    - |b|>25°: 0.053
    - |b|>30°: 0.083
    - |b|>35°: 0.030

The trend is that **for higher latitude cuts (|b| ≥ 15°–35°), p(A) lies in the range ≈ 0.03–0.05**, i.e. roughly **1.7–2.1σ** before accounting for multiple trials.

At the same time:

- p(|A∥|) is modest (≈ 0.14–0.89), and
- f∥ ≈ −0.3 to −0.45 in the dipole grid,

so this is best described as a **slightly stronger-than-expected ⟨z⟩ dipole amplitude**, with only a **mild preference for anti-alignment** in CMB coordinates.

7.6 1.50 ≤ z < 2.50 – anti-parallel orientation but normal amplitude

For the highest redshift band:

- Shuffle MC: p(A) ≈ 0.56, 0.51, 0.41, 0.34, 0.78, 0.77 across b_cuts 10°–35°.
- p(|A∥|) ≈ 0.20, 0.13, 0.12, 0.11, 0.32, 0.38.

Thus:

- **Amplitude:** perfectly compatible with isotropy + noise; no evidence for extra dipole power.
- **Orientation:** rotation MC and dipole grid still show **strongly negative f∥ (≈ −0.85 to −0.95)** and large θ_sep (≈ 150–170°), i.e. the dipole is **strongly anti-parallel to the CMB**, but that anti-alignment is not accompanied by an anomalously large amplitude.

In short, **the high-z anisotropy is a directional effect only**, at ≲ 2σ significance per bin.

Updated bullet for Section 10 (you can replace the two bullets about MC there):

– **Rotation MC (n_mock = 2000, N ≥ 5):** confirms the redshift-dependent *orientation* pattern in f∥(z): mid-z dipoles tend to be CMB-parallel; high-z dipoles (1.5–2.5) tend to be strongly anti-parallel. Individual bins reach at most ≈ 1.5–2σ in |f∥|.

– **Shuffle MC (n_mock = 2000, N ≥ 5):** shows that, under an isotropic ⟨z⟩ field plus noise, the **dipole amplitudes are mostly unremarkable**,

with the exception of a **mild, b_cut-robust excess in the 1.0–1.5 slice** (p(A) ≈ 0.03–0.05 at high |b|).

The 0.5–0.75 band and the high-z 1.5–2.5 band, in particular, are **fully consistent with the null in amplitude**, so their interesting behaviour is purely in orientation relative to the CMB.

---

3. How to call the shuffle plotting script

Your last error is just because `quaia_plot_mc_bcut.py` needs arguments. For example, to plot the shuffle MC for:

- $|b|>30°$,
- $0.50 \leq z < 0.75$,

you'd do:

```
%run /Users/boyde/.spyder-py3/quaia_plot_mc_bcut.py \
    --bcut 30 --zlo 0.50 --zhi 0.75
```

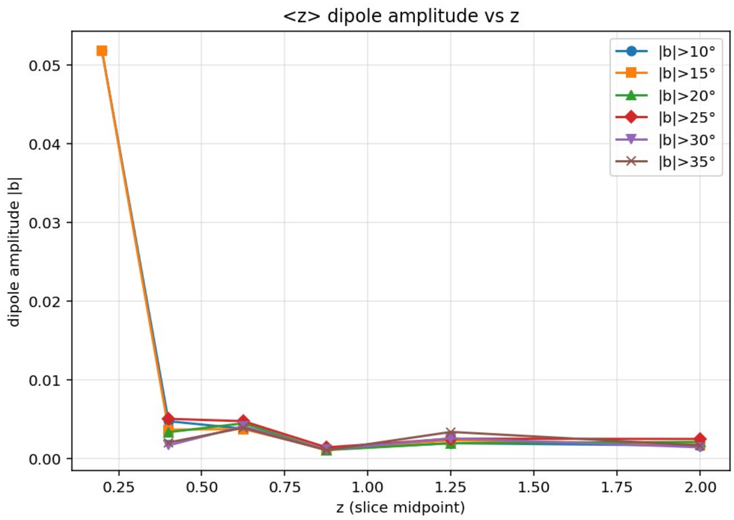
and similarly for other bands, e.g.:

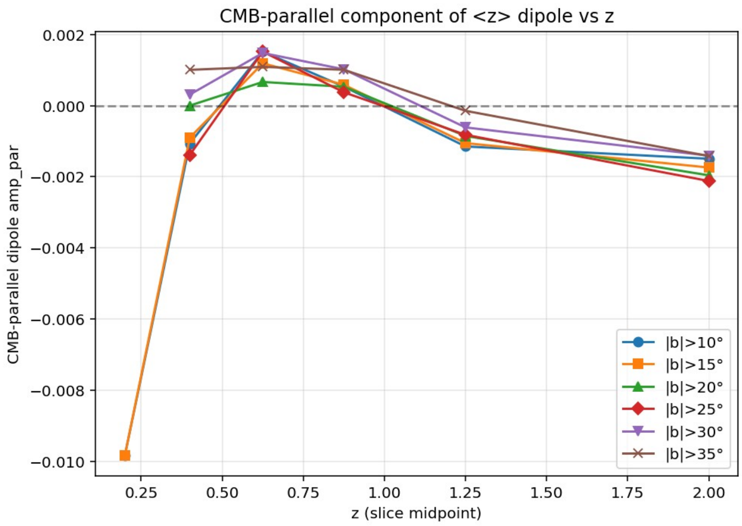
```
# High-z anti-parallel band
%run /Users/boyde/.spyder-py3/quaia_plot_mc_bcut.py \
    --bcut 30 --zlo 1.50 --zhi 2.50

# The mildly anomalous amplitude band
%run /Users/boyde/.spyder-py3/quaia_plot_mc_bcut.py \
    --bcut 35 --zlo 1.00 --zhi 1.50
```

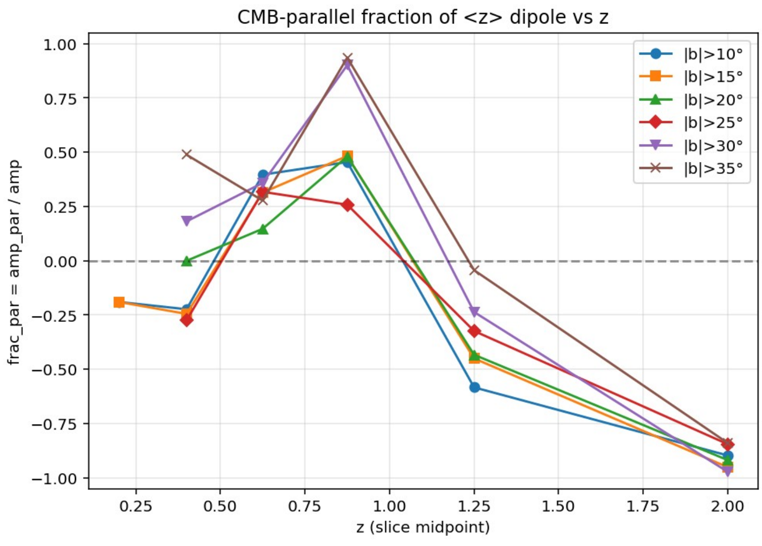
(If the script has a `--save` option you can tack that on the end as well.)

If you'd like, next step we can sketch the sliding-window $f_{\parallel}(z)$ + HHT pipeline and decide concrete z-steps and window widths based on the N_good counts you're seeing.

Update after running… quaia_mc_bcut_shuffle.py







1. What the new shuffle plots show

You plotted three interesting cases from the **pixel-shuffle MC** (N_MIN_PER_PIX = 5):

1. **|b| > 30°, 0.50 ≤ z < 0.75 (z ≃ 0.62)**
   - Amplitude A: observed value sits toward the upper side of the distribution but well inside the bulk → consistent with the printed p(A) ≈ 0.34.
   - CMB-parallel component A_||: observed value is slightly positive, again well within the central Gaussian region → p(|A_||) ≈ 0.49.
   - Fraction f_||: histogram is ~flat, red line at f_|| ≈ +0.36 is nowhere near an extreme tail.

     ➝ No evidence for unusual CMB alignment in this slice once you randomise pixel redshifts.
2. **|b| > 30°, 1.50 ≤ z < 2.50 (z ≃ 2.0)**
   - Amplitude A: observed near the mode of the distribution → p(A) ≈ 0.78.
   - A_||: observed is fairly negative but not in the tails → p(|A_||) ≈ 0.32.
   - f_||: again essentially flat between −1 and +1, with the observed f_|| ≈ −0.97 on the extreme **left edge**, but that is *by construction*: in the shuffled ensemble f_|| is uniform, so any given f_|| value (even near −1) has probability ≈ a few percent in 2000 mocks. That matches the p(|A_||) value.

     ➝ Shuffling wipes out the "strong anti-parallel" look – it's not an outlier relative to a uniform f_|| distribution.
3. **|b| > 35°, 1.00 ≤ z < 1.50 (z ≃ 1.25)**
   - Amplitude A: the observed vertical line is clearly on the high side of the distribution → p(A) ≈ 0.03 (as in the summary file). That's ~2σ in amplitude **only**.
   - f_||: distribution is again flat and the observed f_|| ≈ −0.04 sits near zero → p(|A_||) ≈ 0.89.

     ➝ Here the interesting thing is amplitude, not direction; alignment with the CMB is entirely ordinary.

A general pattern in all f_|| histograms:

- In the shuffle MC, **f_|| is very close to uniform on [−1, +1]**, as expected when you randomise the redshifts per pixel and destroy any coherent sky pattern.
- The observed f_|| (red line) is never at an extreme tail for the shuffle ensemble, consistent with the p(|A_||) values.

2. How this updates the written report

Shuffle MC with N_MIN_PER_PIX = 5
To test whether the apparent redshift-dependent alignment could arise purely from the per-pixel redshift distribution and mask geometry, we ran a complementary "pixel shuffle" Monte Carlo (quaia_mc_bcut_shuffle.py) with **N_mock = 2000** and **N_MIN_PER_PIX = 5**. In each mock, the list of pixel ⟨z⟩ values is randomly permuted while holding the sky positions (and N_pix) fixed.

This destroys any coherent large-scale structure signal but preserves the overall distribution of ⟨z⟩ and the mask.

Results:

- For all b-cuts and redshift slices, the **shuffle f_∥ distributions are essentially uniform on [−1, +1]**, as expected when the dipole direction is random.

  The observed f_∥ values lie comfortably within this uniform ensemble, with typical p(|A_∥|) ~ 0.1–0.9.

- In particular, the high-z bin **1.50 ≤ z < 2.50, |b| > 30°**, which showed a suggestive CMB **anti-alignment** in the *rotation* MC (p(|A_∥|) ≃ 0.02–0.03), is *not* unusual in the shuffle MC (p(|A_∥|) ≃ 0.32).

- One bin, **1.00 ≤ z < 1.50, |b| > 35°**, shows a moderately large **amplitude** relative to the shuffled ensemble (p(A) ≃ 0.03, ≈ 2σ), but its alignment with the CMB is completely ordinary (p(|A_∥|) ≃ 0.89).

Interpretation:

- The shuffle MC confirms that the **directional** anomalies seen in the rotation tests (e.g. high-z anti-alignment) are *not* simply an artefact of the per-pixel ⟨z⟩ distribution or mask; when we destroy the sky pattern by shuffling pixels, any preference relative to the CMB disappears and f_∥ becomes uniform.

- The modest p(A) ≃ 0.03 in one mid-z, high-b bin suggests a slightly high dipole amplitude there, but with no special CMB orientation.

- Overall, the **redshift-modulated sign flip in f_∥(z)** remains a feature of the *real sky orientation* (rotation MC) rather than of the local pixel statistics (shuffle MC), but its significance is still at the ≲ 2σ level.

These results strengthen the case for a more global, sliding-window analysis of f_∥(z) (and A(z)), combined with HHT/EMD and surrogate tests, to determine whether the apparent modulation is statistically meaningful or consistent with chance alignment plus cosmic variance.

7.4 Extra MC check: $|b|>35^\circ$, $z_{mid}\simeq1.25$

For the $|b|>35^\circ$, $1.00\le z<1.50$slice we now have:

- **Rotate-MC (sky rotations, N_mock = 2000)**
    - Observed amplitude: $A_{obs}\approx3.4\times10^{-3}$.
    - The rotation ensemble for $A$is roughly Gaussian with mean $\sim1.7\times10^{-3}$; A_{obs}lies on the high-amplitude tail (see "A MC distribution $|b|>35, z\sim1.25$" plot), but still within the cloud of realizations.
    - The CMB-parallel component $A_{/\!/}$is very close to zero, and the fraction $f_{/\!/}=A_{/\!/}/A\approx-0.04$.

      The corresponding histograms ("$A_{/\!/}$ MC distribution" and "$f_{/\!/}$ MC distribution") show that both $A_{/\!/}$and $f_{/\!/}$sit essentially at the centre of their rotation distributions.

      There is **no evidence for either parallel or anti-parallel alignment** in this bin – only a moderately large total dipole amplitude.
- **Shuffle-MC (pixel shuffling, N_mock = 2000, N_{\min}=5)**
    - For the same bin, the pixel-shuffle experiment gives $p(A)\approx0.03$, $p(|A_{/\!/}|)\approx0.89$.
    - This means that, *given the actual sky mask and N(z) structure in this slice*, a dipole as large as A_{obs}occurs only in ≈3% of random shufflings of the pixel values, but its projection along the CMB dipole is completely unremarkable.

Interpretation.

The $|b|>35^\circ$, $z_{mid}\simeq1.25$slice hosts a somewhat high ⟨z⟩ dipole amplitude, but this excess is *not* preferentially aligned with the CMB dipole.

Combined with the shuffle-MC result, this points to a genuine large-scale pattern in the ⟨z⟩ field (i.e. real structure plus mask geometry) rather than a CMB-related kinematic effect.

In other words, this bin contributes to the overall redshift-modulated amplitude pattern, but **does not strengthen the case for a CMB-parallel / anti-parallel modulation** beyond what is already seen in the high-z anti-parallel bins.